



AEGIS

Anonymization & Exchange Gateway
for Imaging Studies



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A multi-cloud platform for secure, HIPAA-compliant de-identification and sharing of medical imaging data — for research teams and radiology departments alike

In Greek mythology, the aegis was the divine shield of Zeus and Athena — a symbol of protection. The name captures our mission: shielding patient identity while enabling the free flow of imaging data for research and clinical care.

Matthew L. Senjem, M.S.

AEGIS Imaging LLC

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The Problem

Hospitals and research institutions need to share medical imaging data — MRI, CT, PET, ultrasound, X-ray — for multi-site clinical trials, research collaborations, and second opinions. Doing this safely is harder than it should be.

De-identification tools frequently fail. A 2015 study in *European Radiology* tested 10 free DICOM de-identification tools against 50 required PHI data elements. With default settings, only 1 of 10 tools successfully removed all required elements; 4 tools achieved success rates of 26% or less.^[1]

Metadata removal alone is not enough for brain imaging. A 2019 *New England Journal of Medicine* correspondence from Mayo Clinic showed that automated face-recognition software correctly matched de-identified brain MRI participants to their photographs in **83% of cases** — using only the scan geometry, no metadata at all.^[2] This means head imaging requires an additional step: automated defacing (removal of facial surface features from the 3D volume).

Burned-in PHI is invisible to tag-level tools. Patient names, dates of birth, and accession numbers are routinely overlaid directly on image pixels — particularly in ultrasound, CT topograms, and fluoroscopy. Standard DICOM de-identification tools operate on metadata only and miss this entirely.^[3]

Installing software at sending sites is a barrier. Platforms like XNAT require a Java-based desktop client or server daemon installed and IT-approved at every participating hospital. MIRC CTP requires a standalone Java application. Each new site adds months of IT negotiation.

Data sharing is now federally mandated — but tooling lags. The NIH Data Management and Sharing Policy (NOT-OD-21-013, effective January 25, 2023) requires all NIH-funded investigators to share scientific data as a condition of their grant award, no later than the time of first publication.^[4] Institutions are now obligated to share data they have no reliable, low-friction way to de-identify and transmit.

Result: Multi-site studies are delayed by months. In a 2021 study of global randomized trials, contract execution alone averaged 7.9 months per US site.^[5] Software installation and IT approval add to this.

Market Opportunity

The medical image exchange market is growing, driven by federal data sharing mandates and expanding multi-site clinical trial volume in oncology and neurology.

Market	2024 Estimate	2034 Projection	CAGR
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Medical Image Exchange Systems ^[6]	\$3.9B	\$9.9B	9.7%
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Why now:

- The NIH DMS Policy (2023) creates a compliance obligation for tens of thousands of active grants that did not previously require a data sharing plan
- A 2024 NCI-sponsored benchmark (MIDI-B, presented at MICCAI 2024) formally tested de-identification tools and found that burned-in pixel PHI and free-text fields remain the hardest unsolved problems — no existing tool handles all cases reliably^[7].
- Growing multi-site trial volume in oncology (PSMA PET, amyloid PET, whole-body MRI) and neurology (tau PET, fMRI) is driving demand for scalable data collection from distributed hospital networks
- Cloud-hosted DICOM storage (GCS, S3, Azure Blob) and managed databases (Cloud SQL, RDS, Azure Database) are now mature commodity infrastructure, reducing the barrier to building managed platforms

The Solution

AEGIS is a multi-cloud platform that makes secure medical image sharing as easy as uploading a file — for any DICOM modality. It runs on GCP, AWS, or Azure — or locally via Docker Compose. It serves two complementary audiences from a single platform.

Capability	What It Does
No software to install	Upload portal runs entirely in the web browser. Hospitals use their existing computers and internet connection — no desktop app, no IT approval for new software.
Tag-level anonymization in the browser	Patient data is stripped from DICOM tags before it leaves the hospital network, following the DICOM PS3.15 Annex E Basic Confidentiality Profile — the established international standard covering all 18 HIPAA Safe Harbor identifier categories. ^[8]
Automated defacing for head scans	MRI, CT, and PET studies of the head and brain undergo server-side defacing (removal of facial surface geometry) before entering the research archive. Non-head studies bypass this step automatically.
Burned-in PHI detection	Automated OCR scans image pixels for overlaid text (patient names, dates, accession numbers) that tag-level tools miss. Studies with detected pixel PHI

	are flagged for review.
Supports all DICOM modalities	MRI, CT, PET, PET/CT, ultrasound, X-ray, mammography, nuclear medicine, and more — using the same upload and anonymization workflow.
MRI protocol compliance	Automated verification that acquisition parameters (TR, TE, flip angle, resolution) match site-specific templates per scanner manufacturer, model, and software version — catching the 0.19–64% non-compliance rates found across multi-site studies. [10]
Centralized audit trail	Every upload, approval, rejection, and data export is logged with a timestamp and actor. Institutions can demonstrate HIPAA compliance from a single dashboard.
Clinician review before release	Administrators review anonymized images in a web-based DICOM viewer (Weasis DWV) before approving studies for sharing. Side-by-side before/after defacing review is built in. Nothing is shared automatically without human sign-off.
Pixel redaction	Automated detection and masking of burned-in PHI directly in DICOM pixel data — patient names, dates, and accession numbers overlaid on images are located by OCR and redacted at the pixel level.
Neuroimaging analytics	Post-BIDS analysis pipelines with 18 backends: brain (FreeSurfer, SynthSeg, BrainSuite, volBrain, Atlas ROI), spine (SCT, TotalSpineSeg, SPINEPS), whole-body (TotalSegmentator, nnU-Net, MedSAM2, MONAI Label), and specialized (PETSurfer, QSM, BASIL, FSL, ANTs, SPM, ITK-SNAP). Longitudinal analytics (TBM-SyN, FreeSurfer Long) for tracking brain volume changes over time.
Spinal cord analysis	Spinal Cord Toolbox (SCT) integration: automated cord segmentation, cross-sectional area (CSA) per vertebral level, compression metrics (aMCC, aSCOR), and DTI mapping for spinal cord studies.
Comprehensive DICOM format support	JPEG2000 lossless, JPEG-LS, Enhanced (multi-frame) DICOM, and Siemens Mosaic DICOM handled transparently across all processing services via decompression backends.
Vendor private tag preservation	Per-project option to retain vendor-specific private tags (diffusion gradients, CSA headers) during de-identification, with automated PHI scanning of preserved tag values.

Who It's For

AEGIS serves two audiences from a single de-identification and routing platform:

Multi-site research teams need to collect imaging data from hospitals they don't control, de-identify it to meet NIH and IRB requirements, and convert it to research-ready formats. AEGIS gives them a browser link they can send to any participating site — no software installation, no IT negotiation. The research pipeline (protocol compliance, automated QC, BIDS conversion, defacing) ensures data arrives analysis-ready.

Hospital radiology departments need to de-identify imaging data for internal uses — training AI models, building real-world evidence databases, sharing with referring physicians, and fulfilling data requests from payers and registries. AEGIS gives them a centralized de-identification gateway with a complete audit trail, burned-in PHI detection, and configurable routing rules — without replacing their PACS or requiring a separate enterprise integration project.

	Research Teams	Radiology Departments
Entry point	Browser upload link sent to external sites	DIMSE receive / internal batch import
Key need	NIH DMS compliance, multi-site consistency	HIPAA de-identification, audit trail
Differentiator	Zero-install at sending sites	No PACS/VNA integration required
Pipeline features	Defacing, protocol compliance, QC, BIDS	Burned-in PHI, routing rules, export shares
Review workflow	PI/coordinator approves before archive	Radiologist/admin approves before release

The underlying platform — de-identification engine, routing rules, audit trail, Weasis DWV viewer, and admin dashboard — is shared. Research-specific features (defacing, BIDS, protocol templates) and enterprise features (DIMSE receive, HL7/FHIR notifications, tag standardization) extend it for each audience.

How De-identification Works

AEGIS follows a two-phase approach that addresses the limitations documented in the literature:

Phase 1 – In the browser (before upload):

└─ DICOM tags stripped per PS3.15 Annex E Basic Profile [\[8\]](#)

- └ 18 HIPAA Safe Harbor identifiers addressed [\[9\]](#)
- └ Reviewer sees before/after tag comparison before confirming
- └ Only de-identified data is transmitted – PHI never leaves the hospital

Phase 2 – On the server (after upload):

- └ De-identification completeness check
- └ OCR scan of image pixels for burned-in text [\[3\]](#)
- └ Head imaging → automated defacing [\[2\]](#)
- └ Administrator review and approval before data is shared

This two-phase design directly addresses the gaps identified in the Aryanto (2015) [\[1\]](#) and Schwarz (2019) [\[2\]](#) studies.

Supported Modalities

Category	Modalities
Cross-sectional	MRI, CT, PET, PET/CT, PET/MR
Projection	X-Ray (CR/DR), Mammography (FFDM/DBT), Fluoroscopy
Nuclear Medicine	SPECT, Planar Scintigraphy
Ultrasound	Echocardiography, Abdominal, Vascular, OB/GYN
Radiation Therapy	RT Structure, RT Plan, RT Dose, RT Image
Other DICOM	Secondary Capture, Structured Reports, Key Images

Initial focus: Brain MRI, PET, and CT — these exercise the full pipeline including defacing. All other modalities use the same upload and de-identification workflow.

Technology

Layer	Technology	Notes
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Backend	Go on Cloud Run (GCP), ECS Fargate (AWS), or Container Apps (Azure)	Compiled binary, minimal dependencies, fast cold starts
Frontend	React + TypeScript	Runs in any modern browser, no installation required
Database	PostgreSQL 15 (Cloud SQL / RDS)	Audit trail, project/user management, routing rules
DICOM Storage	Cloud-neutral (GCS, S3, Azure Blob, or local filesystem)	Abstracted behind a pluggable storage interface
OCR / PHI Detection	Tesseract OCR (local) / cloud AI (pluggable)	Detects burned-in text in image pixels
Defacing	DeepDefacer (default), mri_deface, mri_reface	Multiple backends with automatic fallback; see docs/research/mri-defacing-tools-comparison.md
DICOM Viewer	Weasis DWV (browser-based)	Lightweight, open-source; supports QIDO-RS/WADO-RS; built-in side-by-side defacing review
Auth	GCP IAP / AWS ALB+Cognito / Azure AD	Multi-provider auth middleware, auto-detection
Processing Pipeline	Microservices architecture — 9 Python microservices (Cloud Run / ECS Fargate / Container Apps) + DIMSE receiver (Compute Engine VM / EC2 / Azure VM) + MCP server (Cloud Run)	Classification, PHI detection + pixel redaction, protocol compliance, QC, defacing, BIDS conversion, synthetic MRI generation, neuroimaging analytics (18 backends incl. FreeSurfer, FSL, ANTs, SPM, SynthSeg, nnU-Net, TotalSegmentator, MONAI Label, PETSurfer, QSM, BASIL), spinal cord analysis (SCT) — auto-dispatched in 4-phase dependency order; DIMSE C-STORE SCP on dedicated VM (static IP, port 11112); MCP server exposes 96 read tools + 95 write tools; agent orchestrator with DICOM tag provenance and diagnostic toolchain
Infrastructure	Terraform (GCP + AWS modules), Docker Compose	Reproducible, version-controlled, multi-cloud; local dev stack starts everything with one command

Live deployments: GCP — [admin.aegisimaging.ai](#) (dashboard) · [api.aegisimaging.ai](#) (API) · AWS — [aws.admin.aegisimaging.ai](#) (dashboard) · [aws.api.aegisimaging.ai](#) (API) · Azure — [azure.admin.aegisimaging.ai](#) (dashboard) · [azure.api.aegisimaging.ai](#) (API).

On the use of automated tools: AEGIS uses automated tools to assist with — not replace — human review. Automated de-identification flags potential issues; a trained administrator reviews and approves every study before it is shared. Automated defacing quality is reviewed side-by-side against the original in the admin interface.

Compliance Framework

AEGIS is designed around established regulatory and technical standards:

Standard	Reference	What It Covers
HIPAA Privacy Rule	45 CFR § 164.514(b)	De-identification methods: Safe Harbor (18 identifiers) and Expert Determination
DICOM Confidentiality Profile	NEMA PS3.15 Annex E	Tag-level de-identification actions for all DICOM attributes
NIH Data Management Policy	NOT-OD-21-013 (eff. Jan 25, 2023)	Data sharing obligations for NIH-funded investigators
Cloud Security	BAA, VPC/VPC-SC, CMEK/KMS	Business Associate Agreement, encrypted storage, network controls — available on GCP, AWS, and Azure

AEGIS does not create a new de-identification standard — it implements the ones that already exist.

Competitive Landscape

	AEGIS	ENCOG (Enlitic)	XNAT	Flywheel	MIRC CTP
Hosting	Multi-cloud (GCP, AWS, Azure)	On-premises / hybrid	Self-hosted	Commercial SaaS	On-premises
Install at sending site	None — browser only	PACS/VNA integration	Desktop Java client	CLI tool	Java application

Burned-in PHI	OCR detection	AI CV detection	None	Built-in	None
Defacing	Automated, reviewable	None	Manual or plugin	Built-in	None
Modalities	All DICOM	MR, CT, XR, US	Neuroimaging focus	All imaging	All imaging
Open source	Private (built on open-source stack)	No (commercial)	Yes	No	Yes
Audit trail	Centralized, per-study	Chain of custody	Per-instance	Built-in	Limited

Existing platforms tend to serve either research (XNAT, Flywheel) or enterprise radiology (ENCOG/Enlitic, MIRC CTP) — but not both. AEGIS is designed for both audiences from a shared platform, and is the only option that runs on any major cloud provider without re-architecture.

Enlitic's ENCOG is the closest commercial analogue to AEGIS's de-identification pipeline.^[12] It uses AI-driven computer vision to detect burned-in text overlays and claims to protect over 4,000 DICOM fields. However, ENCOG requires PACS/VNA integration at each sending site, does not offer volumetric defacing for head imaging, and publicly documents only four modalities (MR, CT, XR, ultrasound). ENCOG is part of a broader commercial suite (Ensign) that includes data standardization (ENDEX, FDA 510(k) cleared) and migration tools — but lacks the research pipeline features (protocol compliance, QC, BIDS) that multi-site studies require.

XNAT and Flywheel serve research well but require software installation at sending sites and offer limited enterprise radiology integration. MIRC CTP is a robust open-source pipeline but requires on-premises Java infrastructure and has no burned-in PHI detection or defacing.

Key differences from existing platforms:

1. **No software installation at sending sites** — the full anonymization workflow runs in the browser. ENCOG, XNAT, and MIRC CTP all require software deployed at the sending institution.
2. **Two-phase de-identification** — both tag-level (browser) and pixel-level (server-side OCR) are addressed in a unified pipeline
3. **Defacing is automated and reviewable** — not optional, manual, or absent. Neither ENCOG nor MIRC CTP offer volumetric defacing.
4. **Not specialty-specific** — designed for all DICOM modalities from the start, unlike XNAT (neuroimaging focus) or ENCOG (4 listed modalities)

5. **Dual-market platform** — research features (protocol compliance, QC, BIDS) and enterprise features (DIMSE receive, routing rules, tag standardization) extend a shared de-identification core. Competitors serve one audience or the other.

Development Velocity

AEGIS was built from a blank repository to full GCP production deployment in **7 days** (February 17–24, 2026), using AI-assisted development tooling. AWS production followed on day 8. Azure Container Apps deployment completed on day 9. The resulting platform is production-grade across all three major clouds: version-controlled infrastructure, automated CI/CD, 2,100+ tests, and all services deployed and monitored simultaneously.

A single merge to **develop** deploys to all three clouds simultaneously — GCP Cloud Build, AWS GitHub Actions, and Azure GitHub Actions (OIDC federated auth) trigger concurrently, updating all services across GCP, AWS, and Azure within minutes from a single shared codebase. Cross-cloud DICOM routing (STOW-RS and DIMSE C-STORE) is verified live across all cloud pairs.

Metric	Value
Days from first commit to GCP production	7
Days from first commit to AWS production	8
Days from first commit to Azure production	9
Git commits	1,300+
API routes (Go)	322+
Automated tests (Go + Python + client)	2,100+
Database migrations	77
Cloud Run services deployed (GCP)	15
ECR repositories provisioned (AWS)	13
ECS Fargate services (AWS)	10

Azure Container Apps	15
Python processing services	9
Pipeline phases	4 (Classification → PHI/Protocol/Deface → QC/BIDS → Analytics/SCT)
MCP AI agent tools (read + write)	191

Phased Roadmap

Production status: The full platform is **live on three clouds**. GCP: Go API, admin dashboard, Weasis DWV viewer, 9 Python processing services (Cloud Run), DIMSE receiver (Compute Engine VM, static IP `35.232.172.221`, port 11112), and MCP server at `api.aegisimaging.ai` and `admin.aegisimaging.ai`. AWS: 10 ECS Fargate services, RDS PostgreSQL, S3, ALB + Cognito auth, DIMSE receiver (EC2 with Elastic IP, port 11112) at `aws.api.aegisimaging.ai` and `aws.admin.aegisimaging.ai`. Azure: 15 Container Apps + PostgreSQL Flexible Server + Azure Blob Storage, DIMSE receiver (Azure Linux VM, static IP `20.97.180.87`, port 11112) at `azure.api.aegisimaging.ai` and `azure.admin.aegisimaging.ai`. All clouds share one codebase; GCP Cloud Build and GitHub Actions deploy in parallel on every merge to `develop`. Cross-cloud DICOM routing (GCP→AWS→Azure) is verified live. **All three DIMSE C-STORE receivers are operational** — accepting inbound studies from PACS systems on TCP port 11112.

✓ Milestone 1 — Foundation + GCP Production (February 17–24, 2026)

Everything listed below was built and deployed to GCP production within 7 days of the first commit:

- Browser-based upload portal with DICOM tag anonymization (PS3.15 Basic Profile, 18 HIPAA identifiers), date shifting, pseudonymization, and before/after tag diff preview
- Go API with 322+ routes: DICOM ingest, routing engine, DICOMweb proxy, export shares, audit trail, webhook subscriptions, API keys, clinical trial access control
- Admin dashboard: study browser, RBAC (admin/viewer + capability-based guards), protocol templates, routing rules, institutions, 17+ management tabs
- Weasis DWV viewer with side-by-side before/after defacing review (yoked scroll synchronization)

- 9 Python processing services on Cloud Run: classification, PHI scan + pixel redaction, protocol compliance, QC, defacing, BIDS conversion, synthetic MRI generation, neuroimaging analytics (18 backends), spinal cord analysis (SCT)
- DIMSE C-STORE SCP on dedicated Compute Engine VM (port 11112) — durable retry queue, dead-letter, exponential backoff
- MCP server (96 read tools + 95 write tools) + agent orchestrator with DICOM tag provenance and diagnostic toolchain — AI-native platform operations
- Terraform IaC (GCP + AWS + Azure modules), GitHub Actions CI, Cloud Build CD (auto-deploy on push to `develop`)
- 2,100+ automated tests (1,072 Go tests, 892 Python pytest tests across 9 sidecars, 166 client library tests)
- Comprehensive DICOM format support: JPEG2000, JPEG-LS, Enhanced multi-frame, Siemens Mosaic
- Vendor private tag preservation with automated PHI scanning of retained tag values
- 8 phases of cross-cloud security hardening with CI enforcement

✓ Milestone 2 — AWS Deployment (February 25, 2026)

- ECS Fargate (10 services), RDS PostgreSQL, S3, ALB + Cognito auth — fully live at `aws.api.aegisimaging.ai`
- Terraform infrastructure provisioned; 13 ECR repositories; GitHub Actions CI/CD auto-deploys on every push to `develop`
- DIMSE receiver on EC2 with Elastic IP, SSM-driven rolling deploys
- **Cross-cloud DICOM routing verified live** — GCP→AWS STOW-RS tested end-to-end; bidirectional API key authentication; routing loop benign (deduplicated by unique constraint)
- Single shared codebase; one merge deploys to all three clouds simultaneously

✓ Milestone 3 — Azure Deployment (February 26, 2026)

- Azure Container Apps deployment — same Go API, Python sidecars, and React frontends as GCP and AWS
- Azure Database for PostgreSQL (Flexible Server), Azure Blob Storage (STORAGE_MODE=azure), Azure Container Registry
- GitHub Actions CI/CD with OIDC federated auth — auto-deploy on every push to `develop`
- DIMSE receiver on Azure Linux VM — same static-IP PACS integration pattern

- Azure Communication Services Email relay for SMTP notifications

✓ Milestone 4 — Advanced Processing & Security (February 27, 2026)

- Comprehensive DICOM format support: JPEG2000 lossless, JPEG-LS, Enhanced multi-frame, Siemens Mosaic across all 9 services
- Pixel redaction pipeline step: automated detection and masking of burned-in PHI in DICOM pixel data
- Vendor private tag preservation with automated PHI scanning of retained tag values
- Clinical trial access control: project membership roles, capability-based write guards, scoped study visibility
- 8 phases of cross-cloud security hardening with CI enforcement scripts

✓ Milestone 5 — Analytics Expansion & MIDI-B Compliance (February 28, 2026)

- Neuroimaging analytics service with 18 backends: brain (FreeSurfer, SynthSeg, BrainSuite, volBrain, Atlas ROI), spine (SCT, TotalSpineSeg, SPINEPS), whole-body (TotalSegmentator, nnU-Net, MedSAM2, MONAI Label), and specialized (PETSURFER, QSM, BASIL, FSL, ANTs, SPM, ITK-SNAP)
- Spinal Cord Toolbox (SCT) sidecar: cord segmentation, cross-sectional area (CSA), compression metrics (aMCC, aSCOR), DTI mapping
- Longitudinal analytics: TBM-SyN tensor-based morphometry and FreeSurfer longitudinal stream for brain volume change tracking
- Biomarker database: ROI volumetric results, QC ratings, demographics, analytics file management
- MIDI-B de-identification benchmark compliance (Tracks 1 & 2): date shifting, pseudonymization, LLM-based text scrubbing, mapping export
- Niivue NIfTI viewer embedded in admin dashboard for analytics output review
- 2,100+ automated tests (1,072 Go, 892 Python, 166 client), 322+ API routes, 191 MCP tools, 77 database migrations

→ Milestone 6 — Private Beta (Q1 2026)

- First enterprise pilot customers (research institutions + radiology departments)
- Invite-gated access with self-service request workflow
- Gathering feedback on de-identification workflows, PACS integration, and protocol compliance

- DIMSE C-STORE receivers operational on all three clouds (GCP, AWS, Azure) — accepting inbound studies from PACS systems on TCP port 11112
- DIMSE C-MOVE / C-FIND workflows for active PACS pull integration
- SLA monitoring, retention policies, and routing rules hardening for production workloads

→ Milestone 7 — Beta Hardening + Compliance Foundations (Q2 2026)

- Business Associate Agreement (BAA) template + countersigning workflow
- SOC 2 Type I audit initiation — point-in-time controls assessment
- DIMSE C-MOVE / C-FIND at scale — AEGIS actively queries and retrieves studies from PACS and VNA systems
- AWS Marketplace listing for enterprise procurement

→ Milestone 8 — SOC 2 Type II + Enterprise Integrations (Q3 2026)

- SOC 2 Type II certification — formal audit after 6-month observation period
- HL7 FHIR notifications — integrate with hospital EMR/RIS systems
- Cross-tenant federated sharing — peer AEGIS instances can exchange approved studies

→ Milestone 9 — Enterprise GA (Q4 2026)

- On-premises deployment option for institutions with strict data residency requirements
- PACS/VNA native query-retrieve — pull studies on demand rather than waiting for push
- Multi-tenant SaaS with per-organization data isolation for radiology groups

Cost Estimate (Development / Proof of Concept)

Resource	GCP Monthly	AWS Monthly	Azure Monthly
Containers (Cloud Run / ECS Fargate / Container Apps)	~\$5–15	~\$5–15	~\$5–15
PostgreSQL (Cloud SQL / RDS / Flexible Server)	~\$10	~\$15	~\$15

Object Storage (GCS / S3 / Azure Blob)	~\$0.02/GB	~\$0.02/GB	~\$0.02/GB
Load Balancer	Included	~\$16 (ALB)	Included
Total (development)	~\$20– 40/month	~\$40– 60/month	~\$25– 50/month

Production costs scale with data volume. A 1,000-session multi-site study (~500 GB) would cost approximately \$50–100/month in storage on any cloud.

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Secure, browser-based medical image de-identification and sharing for any DICOM modality — serving both multi-site research teams and hospital radiology departments from a single multi-cloud platform, with automated defacing, pixel-level PHI detection, and administrator review before any data is released. Runs on GCP, AWS, or Azure.

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